SUBHI TIWARI

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Professional Summary

AI/ML and Python Developer with hands-on experience in Machine Learning, Deep Learning, AI-powered applications, and backend development using FastAPI and Django. Skilled in building intelligent systems and scalable, secure backend services, with a strong foundation in problem-solving, Data Structures, and developing practical software solutions through real-world projects and research.

Experience

Python Developer (Burnett Brücke) Jul 2026 – Present

Developed and enhanced a microservices-based payment processing platform with authentication, authorization, tenant isolation, and fraud-risk evaluation capabilities.

- Implemented an Identity service using FastAPI, supporting OpenID Connect (OIDC) discovery, JWKS, RSA key rotation, and RS256-based access and refresh tokens.
- Secured payment APIs through JWT validation, role-based authorization, tenant isolation, and JWKS caching in the Heimdall API Gateway.
- Contributed to payment workflows, including 3D Secure (3DS), webhook handling, idempotency, and transaction lifecycle management.
- Developed automated tests for authentication, security, concurrency, and failure scenarios.
- Worked with Python, FastAPI, PostgreSQL, Redis, Docker, pytest, and OpenTelemetry in a distributed microservices architecture.

Skills

Programming Languages: Python, C, C++
Backend & Tools: Django, FastAPI, Git, GitHub, VS Code, Jupyter, LangChain, HuggingFace, Docker
Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras, PyTorch
AI & ML: Machine Learning, Deep Learning, NLP, Generative AI, LLMs, RAG, Prompt Engineering
Database: SQL, PostgreSQL
CS Fundamentals: Data Structure & Algorithm, Object-Oriented Programming (OOP), DBMS

Projects

Vedas AI Astrologer Chatbot

- **Technologies: Python, LLM, LangChain, FastAPI, RAG, ChromaDB, Hugging Face**
- Developed an AI-powered chatbot for astrologer-customer interactions.
- Fine-tuned the Qwen2.5 language model on a custom astrology conversation dataset.
- Built an end-to-end LLM pipeline for chat generation, safety validation, and automated quality evaluation.

AI-Based Smart Fertilizer & Pest Detection System

- **Technologies: FastAPI, Python, JavaScript, WeatherAPIs, OpenCV.**
- Built an AI-powered fertilizer recommendation and pest detection using soil data, NPKC values.
- Integrated AI/ML models with YOLOv9 and OpenCV, achieving 90% pest detection accuracy and 93% accuracy in fertilizer prediction.

Experience & Research

ResearchPaper - May 2026

Machine Learning based Rainwater Harvesting Optimization.

- Presented a research paper on a machine learning-based framework for rain water harvesting assessment and cost optimization.
- Developed a prototype integrating machine learning, computer vision, and predictive analysis for sustainability solutions.

Education

INDORE INSTITUTE OF SCIENCE AND TECHNOLOGY
Bachelor of Technology in Computer Science and Engineering | **7.77** **Expected - 2027**

ALPINE PUBLIC SCHOOL
Higher Secondary School | **91%** **2023**

ALPINE PUBLIC SCHOOL
Secondary School | **90%** **2021**

Achievements And Certification

- **AWS – Cloud Foundations –** Training Badge by AWS Academy on March 2026.
- **SQL and RelationalDatabase101** by IBM on Dec 2025.
- **C & C++** Programming.